Online Constrained Control of Storage Systems via Simplex Disturbance-Action Policies

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***Abstract*—We study online control of a scalar storage system with nonnegative adversarial resource arrivals and state-dependent action constraints. The adversarial, time-varying cost depends on both the state and the action. At each time, the controller selects a feasible action before the current resource arrival and cost are revealed. To handle the resulting coupling between feasibility and online learning, we introduce *Simplex Disturbance-Action Control* (SDAC), a policy class specified by H nonnegative parameters whose sum is at most one. The key feature of SDAC is that every fixed policy in the class is feasible by construction. Our online SDAC controller updates these parameters using entropic online mirror descent while preserving feasibility along the time-varying trajectory. We prove a regret bound of $O\Big(\sqrt{T\log(H+1)}\Big)$ relative to the best fixed SDAC policy and a minimax lower bound showing that the dependence on T is tight. Moreover, SDAC geometrically approximates a decreasing-allocation policy class that spreads each arrival over time according to a common decreasing allocation sequence, a structure that arises in stochastic energy-harvesting control. With a suitable choice $H = \Theta(\log T)$, the resulting regret against the decreasing-allocation class is $O\big(\sqrt{T\log\log T}\big)$. The framework applies to energy-harvesting batteries and other storage-constrained resource systems.**

***Index Terms*—Online control, disturbance-action policies, storage systems, energy harvesting, regret minimization, state-dependent action constraints.**

I. Introduction

A. Motivation and Problem Setting

MANY resource-allocation systems have a storage state that changes over time and limits the control action at each time. An action that lowers the current cost may leave less resource for future decisions, while conserving the resource may increase the current cost. The controller must balance these effects while satisfying this constraint at every step.

We study this problem for a scalar storage system that loses a fixed fraction of its stored resource each period. At each time, the controller observes the current state and all past arrivals and cost functions, chooses a control action, and then observes the current arrival and cost function. The arrival and cost sequences may be chosen adversarially, so we do not assume a known probability model. Our goal is to construct an online SDAC controller whose cumulative cost is close to that of the best fixed policy, chosen in hindsight, from a specified benchmark class.

One difficulty is that the current action affects the next state. In online control, *Disturbance-Action Control* (DAC) parameterizes actions using a linear filter of past disturbances to obtain an efficient algorithm with bounded regret. In our storage model, however, some DAC policies can produce negative actions or actions larger than the available resource.

We address this issue by introducing *Simplex Disturbance-Action Control* (SDAC). SDAC restricts the filter parameters to a subprobability simplex and includes powers of the retention coefficient in the filter. These choices ensure that every fixed SDAC policy satisfies the action constraint. The state and action components of each fixed-policy trajectory are also affine in the policy parameter. Thus, a convex state–action cost is convex as a function of that parameter. This structure permits the parameter to be updated using online convex optimization.

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B. Model and Information Pattern

The storage state $x_t \geq 0$ evolves according to

$$x_{t+1} = \alpha x_t - u_t + w_t,$$

where $\alpha \in (0,1)$ is a fixed retention coefficient, u_t is the control action, and $w_t \in [0,1]$ is a nonnegative resource arrival. The action must satisfy

$$0 \leq u_t \leq \alpha x_t.$$

Thus, the controller cannot use more than the amount remaining after storage loss.

At time t, the controller chooses u_t using the current state and the information available through time $t-1$. The arrival w_t and the convex cost function c_t are revealed after the action is selected, and the controller incurs the cost $c_t(x_t, u_t)$. Both the online trajectory and every fixed-policy trajectory used as a benchmark must satisfy the same dynamics and action constraint. The formal model and assumptions are given in Section II.

Our main application is energy-harvesting battery management, where x_t is the battery state of charge, w_t is the harvested energy, u_t is the energy used, and $1-\alpha$ is the fraction of stored energy lost during one period. Similar dynamics arise in perishable inventory systems, water reservoirs with storage loss, and queues with abandonment. In many standard queueing models, $\alpha = 1$; that case is outside our main results and is discussed in Section VII.

C. Contributions

Our contributions are:

- **Policy class.** We introduce SDAC, a finite-memory policy class parameterized by a subprobability simplex. We

prove that every fixed SDAC policy is feasible for every arrival sequence with entries in $[0, 1]$.
We further show that SDAC approximates the decreasing-allocation policy class defined in Section III-C. This class allocates each arrival over future periods according to the same decreasing sequence. It contains every feasible fixed-fraction policy $u_t = kx_t$ for some fixed $k \in [0, \alpha]$, and the approximation error decreases geometrically with the memory horizon H.

- **Online control and regret.** We develop an online SDAC controller that updates its parameter using entropic online mirror descent (OMD) over the subprobability simplex. We prove that the controller remains feasible and achieves $O\Big(\sqrt{T\log(H+1)}\Big)$ regret against the best fixed SDAC policy. For a suitable choice $H = \Theta(\log T)$, the regret against the decreasing-allocation class is $O\big(\sqrt{T\log\log T}\big)$. The same bound holds against the best fixed-fraction policy.
- **Minimax lower bound.** We prove a lower bound of order $\sqrt{T}$ for the expected regret of every causal feasible controller, including randomized controllers; see Theorem 2. When T is at least on the order of $(1-\alpha)^{-1}$, the lower bound scales as $\Omega((1-\alpha)^{-1}\sqrt{T})$ near $\alpha = 1$. The upper and lower bounds differ by a factor of $(1-\alpha)^{-1/2}$ in their dependence on $1-\alpha$. Closing this gap is left for future work.

D. Related Work

a) Online control: Online control is the main basis for this work. Agarwal et al. use DAC policies to obtain $\widetilde{O}(\sqrt{T})$ regret against strongly stable linear controllers [1]. More closely related to our simplex parameterization, Golowich et al. adapt DAC to population dynamics and search over simplex-structured DAC policy parameters [2]. Their state lies in a probability simplex, and their feasible action set is independent of the current state. Here, the constraint $0 \le u_t \le \alpha x_t$ couples each action to the available resource.

Li et al. consider affine constraints on the state and action [3]. Their action constraint is independent of the current state. They enforce feasibility by restricting and projecting their policy updates onto a tightened parameter set. In our work, the action produced by every fixed SDAC policy is feasible by construction. When the parameter changes online, we clip the nominal action to the current admissible interval as in (24).

Liu et al. consider general convex constraint functions [4]. For their adversarial constraints, the guarantees bound cumulative violation and may permit violations at individual times. Such violations are not permitted here, since (2) must hold at every time.

Kumar et al. introduce online convex optimization with unbounded memory [5]. For their discounted infinite-memory model, the lower bound scales as $\sqrt{T}$ divided by one minus the discount factor. This parallels our $\Omega((1-\alpha)^{-1}\sqrt{T})$ lower bound beyond the mixing time, although ours applies directly to causal feasible controllers under the state-dependent action constraint.

b) Energy harvesting: Fixed-fraction and related allocation policies for stochastic energy arrivals motivate our *decreasing-allocation policy class*. Shaviv and Özgür [6] allocate a fixed fraction of the available energy under i.i.d. arrivals. Arafa et al. [7] extend fixed-fraction policies to general concave utilities under i.i.d. arrivals and, for Bernoulli arrivals, characterize a non-increasing allocation sequence following a full recharge. Zibaeenejad and Chen [8] derive an optimal decreasing allocation sequence under Bernoulli arrivals when the controller observes a fixed look-ahead window. Together, these works motivate allocating each arrival over future periods according to a decreasing sequence. Our allocation begins one period after the arrival because the controller acts before observing the current arrival.

Yu and Neely [9] study expected utility maximization under i.i.d. system states, while Asgari and Neely [10] obtain an $O(\sqrt{T})$ regret–battery-capacity tradeoff for arbitrary loss sequences under i.i.d. energy arrivals. Neither objective depends directly on the battery state. Yu and Neely optimize expected time-average utility, whereas Asgari and Neely measure regret against a fixed action. Lin et al. [11] allow arbitrary energy arrivals and time-varying age-of-information objectives. They observe the current arrival before allocation and compare with an unrestricted offline optimum through a competitive ratio. In our setting, the controller acts before observing the current arrival and cost, the arrival and cost sequences may be chosen adversarially, the cost depends on the storage state and action, and the benchmark is a fixed SDAC policy chosen in hindsight.

c) Queueing and network optimization: Classical queueing control studies throughput and queue stability through dynamic service and scheduling decisions [12]–[14], while stochastic network optimization studies tradeoffs between time-average performance and average queue backlog [15]. Our performance criterion is finite-horizon regret against the best fixed SDAC policy in hindsight.

For their time-invariant class, Khojastepour and Sabharwal [16] represent schedulers satisfying a maximum-delay constraint as fixed linear filters of the current and past arrivals, with nonnegative coefficients summing to one rather than at most one as in SDAC. Under i.i.d. arrivals, they identify the uniform moving-average filter as the minimizer of long-run average transmit power over a Gaussian channel. The connection is limited to the simplex-like filter structure: they solve a stochastic average-power problem with a fixed cost, whereas our algorithm updates the SDAC parameters online under adversarial arrival and cost sequences with general convex state–action costs.

d) Inventory and reservoir management: Yang et al. [17] study online procurement under inventory constraints and obtain optimal competitive ratios relative to an offline optimum. They observe the current demand and price before acting, whereas our controller acts before the current arrival and cost are revealed. Hihat and Fermanian [18] learn a base-stock policy for more general inventory dynamics but do not prove a regret bound for their method. For our scalar linear model, we prove feasibility, approximation, and sublinear regret guarantees. Classical stochastic reservoir models study storage under random inflows and fixed release rules [19], with

an emphasis on steady-state behavior rather than finite-horizon online regret.

II. Model and Regret Benchmark

For every positive integer n, we write $[n] \triangleq \{1, \dots, n\}$. For every $z \in \mathbb{R}$, we define $[z]_+ \triangleq \max\{z, 0\}$. For $\mathbf{v} \in \mathbb{R}^n$, the i-th coordinate is denoted by v_i. If the vector already carries a subscript, the coordinate is appended after a comma: the i-th coordinate of $\mathbf{v}_t$ is $v_{t,i}$. We write $\mathbf{1}$ for the all-ones vector, with its dimension clear from context, and $\langle \cdot, \cdot \rangle$ for the Euclidean inner product. We use $\log$ for the natural logarithm.

Fix a positive integer T and a known retention coefficient $\alpha \in (0, 1)$. The state $x_t \in \mathbb{R}_{\geq 0}$ is the resource level at the beginning of time t. The system starts empty:

$$x_1 = 0. \tag{1}$$

At each time $t \in [T]$, the controller chooses an action $u_t \in \mathbb{R}_{\geq 0}$ satisfying

$$0 \leq u_t \leq \alpha x_t. \tag{2}$$

The system then receives an arrival $w_t \in [0, 1]$ and evolves according to

$$x_{t+1} = \alpha x_t - u_t + w_t. \tag{3}$$

Here, $1 - \alpha$ is the fraction of the stored resource lost during one period. The constraint (2) ensures that the action does not exceed the amount remaining after this loss.

In the online-control literature, w_t is commonly called a disturbance. Here, we call it an *arrival* because it is nonnegative, while retaining the standard name Disturbance-Action Control. The nonnegativity of w_t is important for both the storage interpretation and the feasibility properties of SDAC.

At time t, the controller chooses u_t before observing w_t or c_t. After the action is selected, the arrival and the entire cost function are revealed, the controller incurs $c_t(x_t, u_t)$, and the state is updated according to (3).

For every feasible trajectory, (3) and $w_t \in [0, 1]$ give

$$0 \leq x_{t+1} \leq \alpha x_t + 1.$$

Since $x_1 = 0$, induction gives

$$0 \leq x_t \leq \frac{1 - \alpha^{t-1}}{1 - \alpha} \leq \frac{1}{1 - \alpha}, \qquad t \in [T + 1].$$

Together with (2), this bound places every feasible state–action pair in the compact set

$$\mathcal{D}_\alpha \triangleq \left\{ (x, u) : 0 \leq x \leq \frac{1}{1 - \alpha}, 0 \leq u \leq \alpha x \right\}. \tag{4}$$

Remark 1 (Storage capacity). A storage capacity of $1/(1-\alpha)$ suffices to avoid overflow.

Each revealed cost function $c_t : \mathbb{R}^2_{\geq 0} \to \mathbb{R}$ is convex in (x, u) and satisfies the following Lipschitz condition on $\mathcal{D}_\alpha$: there are constants $L_x, L_u > 0$ such that

$$|c_t(x, u) - c_t(x', u')| \leq L_x |x - x'| + L_u |u - u'| \tag{5}$$

for all $(x, u), (x', u') \in \mathcal{D}_\alpha$. We also assume access, at every $(x, u) \in \mathcal{D}_\alpha$, to a cost subgradient

$$\boldsymbol{\zeta}_t(x, u) = \big(\zeta_t^x(x, u), \zeta_t^u(x, u)\big) \in \partial c_t(x, u) \tag{6}$$

satisfying

$$|\zeta_t^x(x, u)| \leq L_x, \qquad |\zeta_t^u(x, u)| \leq L_u.$$

These subgradient bounds hold, for example, when c_t has a convex extension to an open neighborhood of $\mathcal{D}_\alpha$ satisfying (5); see [20, Thm. 9.13].

We use x_t and u_t without parameter arguments for the trajectory generated by the online controller. Its cumulative cost is

$$J_T^{\mathrm{on}} \triangleq \sum_{t=1}^{T} c_t(x_t, u_t). \tag{7}$$

For a fixed positive integer H, let $\pi_{\mathbf{m}}$ denote the fixed SDAC policy indexed by $\mathbf{m} \in \mathcal{M}_H$, where $\mathcal{M}_H$ is defined in Section III. The corresponding *fixed-policy trajectory* is the unique state–action sequence $\{(x_t(\mathbf{m}), u_t(\mathbf{m}))\}_{t=1}^T$ obtained by applying $\pi_{\mathbf{m}}$ at every time. Its cumulative cost is

$$J_T(\mathbf{m}) \triangleq \sum_{t=1}^{T} c_t\big(x_t(\mathbf{m}), u_t(\mathbf{m})\big).$$

The regret against the best fixed SDAC policy is defined by

$$\mathrm{Reg}_T^{\mathrm{SDAC}} \triangleq J_T^{\mathrm{on}} - \min_{\mathbf{m} \in \mathcal{M}_H} J_T(\mathbf{m}). \tag{8}$$

Our objective is to satisfy (2) at every time and achieve sublinear regret.

For the other comparator classes considered below, $x_t(k)$ and $u_t(k)$ denote the state and action induced by the feasible fixed-fraction policy $u_t(k) = k x_t(k)$ for $k \in [0, \alpha]$, while $x_t(\mathbf{q})$ and $u_t(\mathbf{q})$ denote those induced by a decreasing allocation sequence $\mathbf{q}$.

III. Simplex Disturbance-Action Control (SDAC)

A. From Disturbance-Action Control to SDAC

DAC uses a finite linear filter of past disturbances. Under standard conditions, this parameterization yields convex surrogate costs and efficient online updates. In our scalar storage model, DAC has the following form:

$$u_t = \sum_{i=1}^{H} \theta_i w_{t-i}, \tag{9}$$

where H is a positive integer (the memory horizon) and $\boldsymbol{\theta} = (\theta_1, \dots, \theta_H)^\top \in \mathbb{R}^H$ is a linear filter.

For our constrained model, an unrestricted filter does not ensure feasibility: its action can exceed αx_t or become negative. We therefore introduce *Simplex Disturbance-Action Control* (SDAC), which uses a simplex-constrained filter tailored to the storage dynamics. This restriction makes every fixed SDAC policy feasible while keeping surrogate costs convex and online updates efficient.

B. Policy Definition and Structural Properties

Fix a memory horizon $H \in \mathbb{N}_{>0}$ and define the subprobability simplex

$$\mathcal{M}_H \triangleq \left\{\mathbf{m} \in \mathbb{R}^H_{\geq 0} : \|\mathbf{m}\|_1 \leq 1\right\}. \tag{10}$$

With the convention $w_\tau \equiv 0$ for $\tau \leq 0$, each $\mathbf{m} \in \mathcal{M}_H$ defines an SDAC policy $\pi_{\mathbf{m}}$ through

$$u_t(\mathbf{m}) \triangleq \sum_{i=1}^{H} m_i \alpha^i w_{t-i} = \langle \boldsymbol{\phi}^u_t, \mathbf{m} \rangle, \tag{11}$$

where

$$\boldsymbol{\phi}^u_t \triangleq \left(\alpha w_{t-1}, \alpha^2 w_{t-2}, \ldots, \alpha^H w_{t-H}\right)^\top \in \mathbb{R}^H. \tag{12}$$

The resulting SDAC policy class is

$$\Pi^{\mathrm{SDAC}}_H \triangleq \{\pi_{\mathbf{m}} : \mathbf{m} \in \mathcal{M}_H\}. \tag{13}$$

The following state decomposition underlies both structural properties: every $\pi_{\mathbf{m}}$ satisfies the feasibility constraint (2), and its induced cost is convex and Lipschitz in $\mathbf{m}$. Omitted proofs appear in Appendix A.

Lemma 1 (State decomposition under SDAC)**.** *Fix a positive integer H. For $t \in [T]$ and $i \in [H]$, define*

$$r_t(i) \triangleq \sum_{j=1}^{i} \alpha^{j-1} w_{t-j}, \qquad r_t \triangleq \sum_{j=1}^{t-1} \alpha^{j-1} w_{t-j},$$

where the empty sum defining r_1 is zero. Then, for every $\mathbf{m} \in \mathcal{M}_H$ and $t \in [T]$, the fixed-policy state associated with $\mathbf{m}$ satisfies

$$x_t(\mathbf{m}) = \sum_{i=1}^{H} m_i r_t(i) + \big(1 - \sum_{i=1}^{H} m_i\big) r_t. \tag{14}$$

Equivalently, with

$$\boldsymbol{\phi}^x_t \triangleq \big(r_t(1) - r_t, \ldots, r_t(H) - r_t\big)^\top,$$

we have

$$x_t(\mathbf{m}) = r_t + \langle \boldsymbol{\phi}^x_t, \mathbf{m} \rangle.$$

Thus, $\mathbf{m} \mapsto x_t(\mathbf{m})$ is affine.

Lemma 2 (Feasibility of SDAC policies)**.** *For every $\mathbf{m} \in \mathcal{M}_H$, the fixed SDAC policy $\pi_{\mathbf{m}}$ is feasible:*

$$0 \leq u_t(\mathbf{m}) \leq \alpha x_t(\mathbf{m}) \qquad \text{for every } t \in [T].$$

Proof. Fix $t \in [T]$. For every $i \in [H]$,

$$\alpha r_t(i) - \alpha^i w_{t-i} = \sum_{j=1}^{i-1} \alpha^j w_{t-j} \geq 0.$$

Since $\mathbf{m} \in \mathcal{M}_H$, we have $m_i \geq 0$ for every $i \in [H]$ and $1 - \sum_{i=1}^{H} m_i \geq 0$. Also, $w_{t-i} \geq 0$ for every $i \in [H]$. By the definition of r_t and the nonnegativity of the arrivals,

$$r_t = \sum_{j=1}^{t-1} \alpha^{j-1} w_{t-j} \geq 0.$$

Therefore,

$$\begin{aligned} 0 \leq u_t(\mathbf{m}) &= \sum_{i=1}^{H} m_i \alpha^i w_{t-i} \leq \sum_{i=1}^{H} m_i \alpha r_t(i) \\ &\leq \sum_{i=1}^{H} m_i \alpha r_t(i) + \alpha \left(1 - \sum_{i=1}^{H} m_i\right) r_t \\ &= \alpha x_t(\mathbf{m}), \end{aligned}$$

where the last equality follows from (14). □

Lemma 3 (Convexity and Lipschitz continuity of surrogate cost)**.** *For $t \in [T]$, define*

$$\ell_t(\mathbf{m}) \triangleq c_t\big(x_t(\mathbf{m}), u_t(\mathbf{m})\big), \qquad \mathbf{m} \in \mathcal{M}_H,$$

and

$$G_\alpha \triangleq \alpha \left(\frac{L_x}{1-\alpha} + L_u\right). \tag{15}$$

Then ℓ_t is convex on $\mathcal{M}_H$, and, for all $\mathbf{m}, \mathbf{m}' \in \mathcal{M}_H$,

$$|\ell_t(\mathbf{m}) - \ell_t(\mathbf{m}')| \leq G_\alpha \|\mathbf{m} - \mathbf{m}'\|_1.$$

For $\mathbf{m} \in \mathcal{M}_H$, use (6) to define

$$\begin{aligned} (\zeta^x_t, \zeta^u_t) &= \boldsymbol{\zeta}_t\big(x_t(\mathbf{m}), u_t(\mathbf{m})\big), \\ \mathbf{g}_t &\triangleq \zeta^x_t \boldsymbol{\phi}^x_t + \zeta^u_t \boldsymbol{\phi}^u_t. \end{aligned} \tag{16}$$

Then

$$\mathbf{g}_t \in \partial \ell_t(\mathbf{m}), \qquad \|\mathbf{g}_t\|_\infty \leq G_\alpha.$$

C. Approximation of Decreasing-Allocation Policies

Define the set of feasible decreasing allocation sequences

$$\mathcal{Q}_\alpha \triangleq \left\{\mathbf{q} = (q_i)_{i \geq 1} : q_1 \geq q_2 \geq \cdots \geq 0, \quad \sum_{i=1}^{\infty} \frac{q_i}{\alpha^i} \leq 1\right\}. \tag{17}$$

For $\mathbf{q} \in \mathcal{Q}_\alpha$, define

$$\begin{aligned} u_t(\mathbf{q}) &\triangleq \sum_{i=1}^{t-1} q_i w_{t-i}, \\ x_{t+1}(\mathbf{q}) &= \alpha x_t(\mathbf{q}) - u_t(\mathbf{q}) + w_t, \qquad x_1(\mathbf{q}) = 0. \end{aligned} \tag{18}$$

Here, $q_i w_s$ is the amount from arrival w_s allocated i periods later. The restriction $i \geq 1$ gives a one-period delay because w_s is observed only after u_s is chosen. Policies of this form have been studied for Bernoulli recharges [7], [8]. Let

$$\Pi^{\mathrm{alloc}}_\alpha \triangleq \{\pi_{\mathbf{q}} : \mathbf{q} \in \mathcal{Q}_\alpha\}.$$

Lemma 4 (Finite-memory approximation of decreasing-allocation policies)**.** *Every policy in $\Pi^{\mathrm{alloc}}_\alpha$ is feasible for every arrival sequence with entries in $[0,1]$. Fix $\mathbf{q} \in \mathcal{Q}_\alpha$ and an integer $H \geq 1$, and define*

$$m_{H,i}(\mathbf{q}) \triangleq \frac{q_i}{\alpha^i}, \qquad i \in [H].$$

Then $\mathbf{m}_H(\mathbf{q}) \in \mathcal{M}_H$. Let

$$\mathrm{Tail}_H(\mathbf{q}) \triangleq \sum_{i=H+1}^{\infty} q_i,$$

so $\mathrm{Tail}_H(\mathbf{q}) \le \alpha^{H+1}$. *If each* c_t *satisfies* (5)*, then, for every such arrival sequence,*

$$\begin{aligned}\left|\sum_{t=1}^{T}\left[c_t\big(x_t(\mathbf{m}_H(\mathbf{q})),u_t(\mathbf{m}_H(\mathbf{q}))\big)-c_t\big(x_t(\mathbf{q}),u_t(\mathbf{q})\big)\right]\right| \\ \le\left(L_u+\frac{L_x}{1-\alpha}\right)\mathrm{Tail}_H(\mathbf{q})T \\ \le\left(L_u+\frac{L_x}{1-\alpha}\right)\alpha^{H+1}T.\end{aligned} \tag{19}$$

Corollary 1 (Fixed-fraction policies as geometric allocations)**.** *Fix* $k\in[0,\alpha]$*, set* $\lambda_k \triangleq \alpha-k$*, and define*

$$q_i(k)=k\lambda_k^{i-1},\qquad i\ge 1,$$

where $\lambda_k^0 \triangleq 1$*. Then* $\mathbf{q}(k)\in\mathcal{Q}_\alpha$ *and, for every arrival sequence with entries in* $[0,1]$*,*

$$\big(x_t(\mathbf{q}(k)),u_t(\mathbf{q}(k))\big)=\big(x_t(k),u_t(k)\big),\qquad t\in[T].$$

Thus, $\pi_{\mathbf{q}(k)}$ *realizes the feasible fixed-fraction policy* $u_t(k)=kx_t(k)$*. For every integer* $H\ge 1$*, define its SDAC truncation by*

$$\mathbf{m}_H^{\mathrm{frac}}(k)\triangleq\mathbf{m}_H\big(\mathbf{q}(k)\big)\in\mathcal{M}_H,$$

whose omitted tail is

$$\mathrm{Tail}_H\big(\mathbf{q}(k)\big)=\frac{k\lambda_k^H}{1-\lambda_k}.$$

If each c_t *satisfies* (5)*, then*

$$\begin{aligned}\left|\sum_{t=1}^{T}\left[c_t\big(x_t(\mathbf{m}_H^{\mathrm{frac}}(k)),u_t(\mathbf{m}_H^{\mathrm{frac}}(k))\big)-c_t\big(x_t(k),u_t(k)\big)\right]\right| \\ \le\left(L_u+\frac{L_x}{1-\alpha}\right)\frac{k\lambda_k^H}{1-\lambda_k}\,T.\end{aligned} \tag{20}$$

IV. Online SDAC Controller

The controller maintains a parameter $\mathbf{m}_t\in\mathcal{M}_H$ and updates it by OMD with unnormalized negentropy and step size $\eta>0$ [21]:

$$\Psi(\mathbf{m})\triangleq\sum_{i=1}^{H}\big(m_i\log m_i-m_i\big),\qquad \mathbf{m}\in\mathbb{R}^H_{\ge 0}, \tag{21}$$

where $0\log 0\triangleq 0$. For $\mathbf{m}\in\mathbb{R}^H_{\ge 0}$ and $\mathbf{v}\in\mathbb{R}^H_{>0}$, its Bregman divergence is

$$D_\Psi(\mathbf{m}\|\mathbf{v})\triangleq\sum_{i=1}^{H}\Big(m_i\log\tfrac{m_i}{v_i}-m_i+v_i\Big). \tag{22}$$

Initialize the online parameter as

$$\mathbf{m}_1\triangleq\frac{1}{H+1}\mathbf{1}. \tag{23}$$

a) Action selection: At round t, the controller uses the online parameter $\mathbf{m}_t$, initialized by (23) for $t=1$ and produced by the OMD update from round $t-1$ for $t\ge 2$. Fixed-policy feasibility need not hold when $\mathbf{m}_t$ varies: the nominal action $\langle\boldsymbol{\phi}_t^u,\mathbf{m}_t\rangle$ can exceed αx_t. The controller therefore selects

$$u_t\triangleq\min\{\langle\boldsymbol{\phi}_t^u,\mathbf{m}_t\rangle,\alpha x_t\}. \tag{24}$$

The state then evolves according to (3). Since $\boldsymbol{\phi}_t^u$ and $\mathbf{m}_t$ are componentwise nonnegative, $\langle\boldsymbol{\phi}_t^u,\mathbf{m}_t\rangle\ge 0$. If $x_t\ge 0$, then (24) gives $0\le u_t\le\alpha x_t$, and (3) gives $x_{t+1}\ge w_t\ge 0$. Since $x_1=0$, induction shows that the online trajectory is feasible.

Lemma 7 shows that the D_Ψ-projection of $\mathbf{m}_t$ onto the parameters compatible with x_t induces exactly the clipped action u_t.

b) Surrogate cost and OMD update: After w_t and c_t are revealed, the controller evaluates the subgradient $\mathbf{g}_t$ in (16) at $\mathbf{m}_t$ and sets

$$\mathbf{m}_{t+1}\triangleq\operatorname*{arg\,min}_{\mathbf{m}\in\mathcal{M}_H}\left\{\eta\langle\mathbf{g}_t,\mathbf{m}\rangle+D_\Psi(\mathbf{m}\|\mathbf{m}_t)\right\}. \tag{25}$$

Lemma 8 gives a two-step implementation. First, take the coordinatewise multiplicative step

$$\widetilde{m}_{t+1,i}\triangleq m_{t,i}\exp\big(-\eta g_{t,i}\big),\qquad i\in[H]. \tag{26}$$

Next, project onto $\mathcal{M}_H$, which here amounts to rescaling if needed:

$$\begin{aligned}\mathbf{m}_{t+1}&=\operatorname*{arg\,min}_{\mathbf{m}\in\mathcal{M}_H}D_\Psi(\mathbf{m}\|\widetilde{\mathbf{m}}_{t+1})\\&=\frac{\widetilde{\mathbf{m}}_{t+1}}{\max\{1,\|\widetilde{\mathbf{m}}_{t+1}\|_1\}}.\end{aligned} \tag{27}$$

V. Regret Upper Bounds

The resulting online SDAC controller achieves sublinear regret relative to every fixed SDAC policy $\pi_{\mathbf{m}}\in\Pi_H^{\mathrm{SDAC}}$. Omitted proofs appear in Appendix C.

Theorem 1 (Regret against fixed SDAC policies)**.** *Suppose the costs satisfy the assumptions of Section II. For any step size* $\eta>0$*, the controller in Section IV, initialized by* (23)*, satisfies*

$$\mathrm{Reg}_T^{\mathrm{SDAC}}\le\frac{\log(H+1)}{\eta}+\eta T\Gamma_\alpha, \tag{28}$$

where

$$\Gamma_\alpha\triangleq\frac{G_\alpha^2}{2}+\frac{\alpha G_\alpha(L_x+\alpha L_u)}{(1-\alpha)^2}, \tag{29}$$

and $G_\alpha=\alpha\big(\frac{L_x}{1-\alpha}+L_u\big)$ *as in* (15)*. Choosing*

$$\eta^*=\sqrt{\frac{\log(H+1)}{T\Gamma_\alpha}}$$

gives

$$\mathrm{Reg}_T^{\mathrm{SDAC}}\le 2\sqrt{\Gamma_\alpha T\log(H+1)}.$$

For fixed $L_x,L_u>0$*,*

$$\sqrt{\Gamma_\alpha}=\Theta\Big((1-\alpha)^{-3/2}\Big)\quad\text{as }\alpha\uparrow 1.$$

The proof combines the OMD bound for the surrogate costs (Lemma 10) with the following bound on the difference between realized and surrogate costs.

Lemma 5 (Stability under online parameter updates)**.** *Under the assumptions of Theorem 1, the online SDAC controller satisfies*

$$\sum_{t=1}^{T}\Big(c_t(x_t,u_t)-\ell_t(\mathbf{m}_t)\Big)\le\frac{\alpha\eta TG_\alpha(L_x+\alpha L_u)}{(1-\alpha)^2}. \tag{30}$$

Corollary 2 (No-regret against decreasing-allocation policies)**.** *Under the assumptions of Theorem 1, run the online SDAC controller with step size η^* and memory horizon*

$$H=H_T\triangleq\max\left\{1,\left\lceil\frac{\log T}{\log(1/\alpha)}\right\rceil\right\},$$

where η^* is given in that theorem. Define

$$J_T(\mathbf{q})\triangleq\sum_{t=1}^{T}c_t\big(x_t(\mathbf{q}),u_t(\mathbf{q})\big),\qquad\mathbf{q}\in\mathcal{Q}_\alpha,$$
$$\mathrm{Reg}_T^{\mathrm{alloc}}\triangleq J_T^{\mathrm{on}}-\inf_{\mathbf{q}\in\mathcal{Q}_\alpha}J_T(\mathbf{q}).$$

Then

$$\mathrm{Reg}_T^{\mathrm{alloc}}\le 2\sqrt{\Gamma_\alpha T\log(H_T+1)}+\left(L_u+\frac{L_x}{1-\alpha}\right)\alpha^{H_T+1}T. \tag{31}$$

For fixed α, L_x, L_u, the right-hand side is $O\big(\sqrt{T\log\log T}\big)=o(T)$.

The controller also satisfies a regret bound against fixed-fraction policies because geometric allocation sequences induce these policies.

Corollary 3 (No-regret against fixed-fraction policies)**.** *Run the online SDAC controller as in Corollary 2. For $k\in[0,\alpha]$, let $J_T(k)\triangleq\sum_{t=1}^{T}c_t\big(x_t(k),u_t(k)\big)$, where $u_t(k)=kx_t(k)$, and define*

$$\mathrm{Reg}_T^{\mathrm{frac}}\triangleq J_T^{\mathrm{on}}-\min_{k\in[0,\alpha]}J_T(k).$$

Then (31) *also holds with* $\mathrm{Reg}_T^{\mathrm{alloc}}$ *replaced by* $\mathrm{Reg}_T^{\mathrm{frac}}$.

VI. Minimax Regret Lower Bound

Complementing the upper bound, the next result shows that the $\sqrt{T}$ dependence is unavoidable and quantifies the necessary dependence on $1-\alpha$ for any causal feasible controller. Omitted proofs appear in Appendix D.

Theorem 2 (Minimax SDAC regret lower bound)**.** *Fix $H\ge1$ and $T\ge2$. For every causal feasible online controller $\mathcal{A}$, possibly randomized, there is a deterministic oblivious sequence of arrivals and costs satisfying the assumptions of Section II such that*

$$\mathbb{E}_{\mathcal{A}}\big[\mathrm{Reg}_T^{\mathrm{SDAC}}\big]\ge\frac{1}{2\sqrt{2}}\sqrt{\sum_{t=2}^{T}\beta_t^2}, \tag{32}$$

where

$$\beta_t\triangleq\alpha\left(L_u+\frac{L_x(1-\alpha^{t-2})}{1-\alpha}\right),\qquad t=2,\dots,T. \tag{33}$$

The expectation is over the internal randomization of $\mathcal{A}$.

Corollary 4 (Lower bound beyond the mixing time)**.** *Fix $H\ge1$ and define*

$$t_{\mathrm{mix}}(\alpha)\triangleq\left\lceil\frac{\log 2}{\log(1/\alpha)}\right\rceil.$$

If $T\ge2t_{\mathrm{mix}}(\alpha)+2$, then for every causal feasible controller $\mathcal{A}$, possibly randomized, there is a deterministic oblivious sequence satisfying the model assumptions such that

$$\mathbb{E}_{\mathcal{A}}\big[\mathrm{Reg}_T^{\mathrm{SDAC}}\big]\ge\frac{G_\alpha}{8}\sqrt{T},$$

where $G_\alpha=\alpha(L_x/(1-\alpha)+L_u)$ as in (15)*.*

Remark 2 (Large-α regime). For fixed $L_x,L_u>0$, both $t_{\mathrm{mix}}(\alpha)$ and G_α are of order $(1-\alpha)^{-1}$ as $\alpha\uparrow1$. If $T\ge2t_{\mathrm{mix}}(\alpha)+2$, then Corollary 4 yields

$$\mathbb{E}_{\mathcal{A}}\big[\mathrm{Reg}_T^{\mathrm{SDAC}}\big]=\Omega\left(\frac{\sqrt{T}}{1-\alpha}\right).$$

The upper bound is $O((1-\alpha)^{-3/2}\sqrt{T\log(H+1)})$ by Theorem 1. Its dependence on $(1-\alpha)^{-1}$ is therefore larger by a factor of $(1-\alpha)^{-1/2}$.

This scaling requires T to grow with $t_{\mathrm{mix}}(\alpha)$. For fixed T,

$$\frac{1-\alpha^{t-2}}{1-\alpha}\longrightarrow t-2\qquad\text{as }\alpha\uparrow1,$$

so the lower bound in (32) remains finite.

VII. Discussion and Extensions

We briefly discuss three extensions. In each case, the feasibility, convexity, and stability arguments used above continue to apply after the corresponding changes to the policy domain and the Lipschitz constants. We state these changes but omit the resulting regret bounds, whose form is unchanged up to the modified constants and, in the vector settings, the chosen product norm.

a) Retention $\alpha\ge1$: For $\alpha\ge1$, fix k such that $\bar\alpha\triangleq\alpha-k\in(0,1)$ and write $u_t=kx_t+v_t$. Then

$$x_{t+1}=\bar\alpha x_t-v_t+w_t.$$

An SDAC filter for v_t uses the powers $\bar\alpha^i$. If $0\le v_t\le\bar\alpha x_t$, then

$$kx_t\le u_t=kx_t+v_t\le(k+\bar\alpha)x_t=\alpha x_t,$$

so the original feasibility constraint (2) is satisfied. Since

$$x_{t+1}\le\bar\alpha x_t+1,\qquad x_1=0,$$

induction gives $x_t\le1/(1-\bar\alpha)$. The transformed cost $c_t(x,kx+v)$ remains convex in (x,v). Thus the analysis applies when the costs satisfy the corresponding Lipschitz and bounded-subgradient assumptions on the bounded domain

$$0\le x\le\frac{1}{1-\bar\alpha},\qquad kx\le u\le\alpha x.$$

For a fixed k, the comparator class is correspondingly restricted to policies whose actions satisfy $u_t\ge kx_t$. Selecting k, either in advance or online, is left for future work.

b) Vector actions and multi-task allocation: Consider a single storage state x_t with a vector action $\mathbf{u}_t \in \mathbb{R}^n_{\geq 0}$ that allocates resources across n tasks, subject to $\sum_{a=1}^n u_{t,a} \leq \alpha x_t$. Thus, the total drain from storage is the sum of the task allocations. The dynamics remain $x_{t+1} = \alpha x_t - \sum_{a=1}^n u_{t,a} + w_t$, and the cost $c_t(x_t, \mathbf{u}_t)$ is convex in $(x, \mathbf{u})$. Replace $\mathbf{m} \in \mathcal{M}_H$ with a matrix $\mathbf{M} \in \mathbb{R}^{n \times H}_{\geq 0}$ satisfying

$$\sum_{a=1}^{n}\sum_{i=1}^{H} M_{a,i} \leq 1, \qquad u_{t,a}(\mathbf{M}) = \sum_{i=1}^{H} M_{a,i}\alpha^i w_{t-i}.$$

Set $m_i = \sum_{a=1}^n M_{a,i}$. Then

$$m_i \geq 0, \qquad \sum_{i=1}^{H} m_i = \sum_{a=1}^{n}\sum_{i=1}^{H} M_{a,i} \leq 1,$$

so $\mathbf{m} \in \mathcal{M}_H$. Moreover,

$$\sum_{a=1}^{n} u_{t,a}(\mathbf{M}) = \sum_{i=1}^{H} m_i \alpha^i w_{t-i} \leq \alpha x_t(\mathbf{M}),$$

where the inequality is the scalar SDAC feasibility bound. Both the state and vector action are affine in $\mathbf{M}$. Therefore, $c_t\big(x_t(\mathbf{M}), \mathbf{u}_t(\mathbf{M})\big)$ is convex in $\mathbf{M}$. For the online SDAC controller, a nominal vector that exceeds the available total resource can be scaled proportionally so that its entries remain nonnegative and sum to αx_t. Under an ℓ_1 Lipschitz bound in the action, the size of this correction equals the excess total action, and the stability proof proceeds with the corresponding matrix norm. OMD then runs over the matrix subprobability simplex above.

c) Multiple storage states with diagonal dynamics: Suppose the state is $\mathbf{x}_t \in \mathbb{R}^d_{\geq 0}$ and evolves according to $\mathbf{x}_{t+1} = \mathbf{A}\mathbf{x}_t - \mathbf{u}_t + \mathbf{w}_t$, where $\mathbf{A} = \operatorname{diag}(\alpha_1, \ldots, \alpha_d)$, $\alpha_s \in (0,1)$ for $s \in [d]$, the arrivals satisfy $\mathbf{w}_t \in [0,1]^d$, and the per-storage constraints are $0 \leq u_{t,s} \leq \alpha_s x_{t,s}$. Assign one SDAC parameter in $\mathcal{M}_H$ to each storage state and use the powers α_s^i in coordinate s. The dynamics and feasibility constraints then decouple by storage, and the SDAC parameter domain is a product of subprobability simplices, with $x_{t,s} \leq 1/(1-\alpha_s)$ for every $s \in [d]$. The online SDAC controller clips each nominal action to $\alpha_s x_{t,s}$. If the cost is jointly convex and satisfies compatible Lipschitz and bounded-subgradient assumptions on the resulting bounded product domain, the surrogate cost is jointly convex in all SDAC parameters and OMD applies on the product set. With a sum of negentropy regularizers, the OMD minimization separates by storage, although each subgradient may depend on all storage states and actions. Cross-coupled storage systems (nondiagonal $\mathbf{A}$) require further analysis.

VIII. Conclusion

We studied adversarial online control of a storage system under a state-dependent action constraint. We introduced SDAC so that every fixed-policy trajectory is feasible and developed an online SDAC controller with $O\Big(\sqrt{T\log(H+1)}\Big)$ regret against the best fixed SDAC policy. For $T \geq 2t_{\mathrm{mix}}(\alpha) + 2$, the minimax lower bound gives $\Omega((1-\alpha)^{-1}\sqrt{T})$ regret as $\alpha \uparrow 1$; hence, a factor of $(1-\alpha)^{-1/2}$ between the upper and lower bounds remains open. The SDAC approximation results further give $O\big(\sqrt{T\log\log T}\big)$ regret against the best decreasing-allocation policy for a suitable choice $H = \Theta(\log T)$. The model applies directly to energy-harvesting battery management and related settings in which storage limits must be respected in real time.

Appendix A
Proofs for Simplex Disturbance-Action Control

Proof of Lemma 1. For $1 \leq t < T$ and $i \in [H]$, the definitions give

$$r_{t+1} = \sum_{j=1}^{t} \alpha^{j-1} w_{t+1-j} = w_t + \sum_{j=1}^{t-1} \alpha^j w_{t-j} = w_t + \alpha r_t,$$

$$\begin{aligned} r_{t+1}(i) &= \sum_{j=1}^{i} \alpha^{j-1} w_{t+1-j} \\ &= w_t + \sum_{j=1}^{i-1} \alpha^j w_{t-j} = w_t + \alpha r_t(i) - \alpha^i w_{t-i}. \end{aligned}$$

The sums with upper limits $t-1$ and $i-1$ are empty when $t=1$ and $i=1$, respectively.

At $t=1$, (1) gives $x_1(\mathbf{m}) = 0$, while $r_1(i) = r_1 = 0$ by the arrival and empty-sum conventions. Thus, (14) holds at $t=1$. Suppose it holds at some t with $1 \leq t < T$. By (3) and (11),

$$\begin{aligned} x_{t+1}(\mathbf{m}) &= \alpha x_t(\mathbf{m}) - u_t(\mathbf{m}) + w_t \\ &= \alpha\left[\sum_{i=1}^{H} m_i r_t(i) + \left(1 - \sum_{i=1}^{H} m_i\right) r_t\right] \\ &\quad - \sum_{i=1}^{H} m_i \alpha^i w_{t-i} + w_t \\ &= \sum_{i=1}^{H} m_i\big(\alpha r_t(i) - \alpha^i w_{t-i}\big) \\ &\quad + \left(1 - \sum_{i=1}^{H} m_i\right)\alpha r_t + w_t \\ &= \sum_{i=1}^{H} m_i\big(r_{t+1}(i) - w_t\big) \\ &\quad + \left(1 - \sum_{i=1}^{H} m_i\right)(r_{t+1} - w_t) + w_t \\ &= \sum_{i=1}^{H} m_i r_{t+1}(i) + \left(1 - \sum_{i=1}^{H} m_i\right) r_{t+1}. \end{aligned}$$

Thus, (14) follows by induction. □

Proof of Lemma 3. Lemma 1 and (11) give

$$x_t(\mathbf{m}) = r_t + \langle \boldsymbol{\phi}_t^x, \mathbf{m}\rangle, \qquad u_t(\mathbf{m}) = \langle \boldsymbol{\phi}_t^u, \mathbf{m}\rangle. \tag{34}$$

For $i \in [H]$, Lemma 1 and the definitions of $r_t(i)$ and r_t give

$$|\phi^x_{t,i}| = r_t - r_t(i) = \sum_{j=i+1}^{t-1} \alpha^{j-1} w_{t-j}$$
$$\le \sum_{j=i+1}^{\infty} \alpha^{j-1} = \frac{\alpha^i}{1-\alpha} \le \frac{\alpha}{1-\alpha},$$
$$0 \le \phi^u_{t,i} = \alpha^i w_{t-i} \le \alpha^i \le \alpha.$$

The sum is empty when $i \ge t-1$. Taking the maximum over $i \in [H]$ gives

$$\|\boldsymbol{\phi}^x_t\|_\infty \le \frac{\alpha}{1-\alpha}, \qquad \|\boldsymbol{\phi}^u_t\|_\infty \le \alpha. \tag{35}$$

Since the state–action pair in (34) is affine in $\mathbf{m}$ and c_t is convex, ℓ_t is convex.

By Lemma 2, the fixed-policy state–action pairs associated with $\mathbf{m}$ and $\mathbf{m}'$ belong to $\mathcal{D}_\alpha$. Thus, (5), (34), Hölder's inequality, and (35) give

$$\begin{aligned}|\ell_t(\mathbf{m}) - \ell_t(\mathbf{m}')| &\le L_x|x_t(\mathbf{m}) - x_t(\mathbf{m}')| \\ &\quad + L_u|u_t(\mathbf{m}) - u_t(\mathbf{m}')| \\ &= L_x\big|\langle \boldsymbol{\phi}^x_t, \mathbf{m} - \mathbf{m}'\rangle\big| \\ &\quad + L_u\big|\langle \boldsymbol{\phi}^u_t, \mathbf{m} - \mathbf{m}'\rangle\big| \\ &\le \left(\frac{\alpha L_x}{1-\alpha} + \alpha L_u\right)\|\mathbf{m} - \mathbf{m}'\|_1 \\ &= G_\alpha \|\mathbf{m} - \mathbf{m}'\|_1.\end{aligned}$$

Fix $\mathbf{m} \in \mathcal{M}_H$, and let (ζ^x_t, ζ^u_t) and $\mathbf{g}_t$ be defined by (16). For every $\mathbf{m}' \in \mathcal{M}_H$, the subgradient inequality gives

$$\begin{aligned}\ell_t(\mathbf{m}') &= c_t\big(x_t(\mathbf{m}'), u_t(\mathbf{m}')\big) \ge c_t\big(x_t(\mathbf{m}), u_t(\mathbf{m})\big) \\ &\quad + \zeta^x_t\big(x_t(\mathbf{m}') - x_t(\mathbf{m})\big) + \zeta^u_t\big(u_t(\mathbf{m}') - u_t(\mathbf{m})\big) \\ &= \ell_t(\mathbf{m}) + \zeta^x_t\langle \boldsymbol{\phi}^x_t, \mathbf{m}' - \mathbf{m}\rangle + \zeta^u_t\langle \boldsymbol{\phi}^u_t, \mathbf{m}' - \mathbf{m}\rangle \\ &= \ell_t(\mathbf{m}) + \langle \mathbf{g}_t, \mathbf{m}' - \mathbf{m}\rangle.\end{aligned}$$

Thus $\mathbf{g}_t \in \partial\ell_t(\mathbf{m})$. The bounded-subgradient assumption and (35) give

$$\|\mathbf{g}_t\|_\infty \le |\zeta^x_t|\|\boldsymbol{\phi}^x_t\|_\infty + |\zeta^u_t|\|\boldsymbol{\phi}^u_t\|_\infty \le \frac{\alpha L_x}{1-\alpha} + \alpha L_u$$
$$= G_\alpha. \qquad \square$$

Proof of Lemma 4. Since $\mathbf{q} \in \mathcal{Q}_\alpha$,

$$m_{H,i}(\mathbf{q}) = \frac{q_i}{\alpha^i} \ge 0, \qquad \|\mathbf{m}_H(\mathbf{q})\|_1 = \sum_{i=1}^{H} \frac{q_i}{\alpha^i} \le 1.$$

Thus $\mathbf{m}_H(\mathbf{q}) \in \mathcal{M}_H$.

For feasibility, use truncation length T. Since $w_\tau = 0$ for $\tau \le 0$, for every $t \in [T]$,

$$u_t\big(\mathbf{m}_T(\mathbf{q})\big) = \sum_{i=1}^{T} \frac{q_i}{\alpha^i}\alpha^i w_{t-i} = \sum_{i=1}^{t-1} q_i w_{t-i} = u_t(\mathbf{q}).$$

The two policies have the same initial state and state recursion, so their state trajectories coincide. Lemma 2 therefore gives

$$0 \le u_t(\mathbf{q}) \le \alpha x_t(\mathbf{q}), \qquad t \in [T].$$

For the approximation error, define

$$e^u_t \triangleq u_t(\mathbf{q}) - u_t\big(\mathbf{m}_H(\mathbf{q})\big), \qquad e^x_t \triangleq x_t\big(\mathbf{m}_H(\mathbf{q})\big) - x_t(\mathbf{q}).$$

Again using $w_\tau = 0$ for $\tau \le 0$,

$$e^u_t = \sum_{i=H+1}^{\infty} q_i w_{t-i}, \qquad 0 \le e^u_t \le \mathrm{Tail}_H(\mathbf{q}).$$

Subtracting the two state recursions gives

$$e^x_{t+1} = \alpha e^x_t + e^u_t, \qquad e^x_1 = 0.$$

Hence, with the sum understood to be zero at $t = 1$,

$$0 \le e^x_t = \sum_{s=1}^{t-1} \alpha^{t-1-s} e^u_s \le \mathrm{Tail}_H(\mathbf{q}) \sum_{s=1}^{t-1} \alpha^{t-1-s}$$
$$= \mathrm{Tail}_H(\mathbf{q}) \frac{1-\alpha^{t-1}}{1-\alpha} \le \frac{\mathrm{Tail}_H(\mathbf{q})}{1-\alpha}.$$

Both policies are feasible, so their state–action pairs belong to $\mathcal{D}_\alpha$. By (5) and the triangle inequality,

$$\left|\sum_{t=1}^{T}\big[c_t\big(x_t(\mathbf{m}_H(\mathbf{q})), u_t(\mathbf{m}_H(\mathbf{q}))\big) - c_t\big(x_t(\mathbf{q}), u_t(\mathbf{q})\big)\big]\right|$$
$$\le \sum_{t=1}^{T}(L_x e^x_t + L_u e^u_t) \le \left(L_u + \frac{L_x}{1-\alpha}\right)\mathrm{Tail}_H(\mathbf{q})T.$$

Finally, since $\alpha \in (0,1)$,

$$\mathrm{Tail}_H(\mathbf{q}) = \sum_{i=H+1}^{\infty} \alpha^i \frac{q_i}{\alpha^i} \le \alpha^{H+1} \sum_{i=H+1}^{\infty} \frac{q_i}{\alpha^i} \le \alpha^{H+1}.$$

This proves (19). $\square$

Proof of Corollary 1. Since $0 \le \lambda_k \le \alpha < 1$,

$$q_i(k) \ge 0, \qquad q_{i+1}(k) = \lambda_k q_i(k) \le q_i(k), \qquad i \ge 1.$$

If $k = 0$, then $\mathbf{q}(k) = \mathbf{0} \in \mathcal{Q}_\alpha$. If $k > 0$, then $0 \le \lambda_k/\alpha < 1$ and

$$\sum_{i=1}^{\infty} \frac{q_i(k)}{\alpha^i} = \frac{k}{\alpha}\sum_{j=0}^{\infty}\left(\frac{\lambda_k}{\alpha}\right)^j = \frac{k}{\alpha}\frac{1}{1-\lambda_k/\alpha}$$
$$= \frac{k}{\alpha - \lambda_k} = 1.$$

Thus, $\mathbf{q}(k) \in \mathcal{Q}_\alpha$.

Under the fixed-fraction policy, (3) becomes

$$x_{t+1}(k) = \lambda_k x_t(k) + w_t.$$

Iterating this recursion from $x_1(k) = 0$ gives

$$x_t(k) = \sum_{i=1}^{t-1} \lambda_k^{i-1} w_{t-i},$$
$$u_t(k) = k x_t(k) = \sum_{i=1}^{t-1} q_i(k) w_{t-i} = u_t\big(\mathbf{q}(k)\big).$$

The two policies have the same actions, initial state, and dynamics. Hence, their state trajectories coincide. Lemma 4 gives $\mathbf{m}^{\mathrm{frac}}_H(k) \in \mathcal{M}_H$. Moreover,

$$\mathrm{Tail}_H\big(\mathbf{q}(k)\big) = k\sum_{i=H+1}^{\infty}\lambda_k^{i-1} = k\lambda_k^H\sum_{j=0}^{\infty}\lambda_k^j$$
$$= \frac{k\lambda_k^H}{1-\lambda_k}.$$

Substituting the trajectory and tail identities into (19) proves (20). $\square$

APPENDIX B
PROOFS FOR THE ONLINE SDAC CONTROLLER

The probability-simplex results for OMD with unnormalized negentropy and its Bregman divergence are standard [21]. We state and prove the corresponding forms used below because the online parameter set $\mathcal{M}_H$ is a subprobability simplex.

Lemma 6 (Strong convexity of negentropy on the subprobability simplex)**.** *For every $\mathbf{m} \in \mathcal{M}_H$ and $\mathbf{v} \in \mathcal{M}_H \cap \mathbb{R}^H_{>0}$,*

$$D_\Psi(\mathbf{m}\|\mathbf{v}) \geq \frac{1}{2}\|\mathbf{m}-\mathbf{v}\|_1^2. \tag{36}$$

Thus, Ψ is 1-strongly convex with respect to $\|\cdot\|_1$ on $\mathcal{M}_H$.

Proof. First suppose $\mathbf{m} > 0$ coordinatewise. Set $\mathbf{d} \triangleq \mathbf{m}-\mathbf{v}$ and $\mathbf{z}_s \triangleq \mathbf{v} + s\mathbf{d}$ for $s \in [0,1]$. Then $\mathbf{z}_s \in \mathcal{M}_H \cap \mathbb{R}^H_{>0}$, and weighted Cauchy–Schwarz gives

$$\begin{aligned}\|\mathbf{d}\|_1^2 &= \left(\sum_{i=1}^H \frac{|d_i|}{\sqrt{z_{s,i}}}\sqrt{z_{s,i}}\right)^2 \leq \left(\sum_{i=1}^H \frac{d_i^2}{z_{s,i}}\right)\left(\sum_{i=1}^H z_{s,i}\right)\\ &\leq \mathbf{d}^\top \nabla^2\Psi(\mathbf{z}_s)\mathbf{d}.\end{aligned}$$

The integral remainder in Taylor's formula therefore yields

$$\begin{aligned}D_\Psi(\mathbf{m}\|\mathbf{v}) &= \int_0^1 (1-s)\mathbf{d}^\top\nabla^2\Psi(\mathbf{z}_s)\mathbf{d}\,ds\\ &\geq \|\mathbf{d}\|_1^2\int_0^1 (1-s)\,ds = \frac{1}{2}\|\mathbf{d}\|_1^2.\end{aligned}$$

For a general $\mathbf{m} \in \mathcal{M}_H$ and $0 < \varepsilon < 1$, apply the bound to $\mathbf{m}^\varepsilon \triangleq (1-\varepsilon)\mathbf{m} + \varepsilon\mathbf{v}$ and let $\varepsilon \downarrow 0$. The result follows from the convention $0\log 0 = 0$ and continuity in the first argument of D_Ψ. □

Lemma 7 (Action clipping as a Bregman projection)**.** *For $t \in [T]$, define the set of parameters compatible with the current state by*

$$\mathcal{B}_t \triangleq \{\mathbf{m} \in \mathcal{M}_H : \langle \boldsymbol{\phi}_t^u, \mathbf{m}\rangle \leq \alpha x_t\}.$$

Let $\mathbf{m}_t^{\mathrm{exec}}$ be the Bregman projection of $\mathbf{m}_t$ onto $\mathcal{B}_t$:

$$\mathbf{m}_t^{\mathrm{exec}} \in \operatorname*{arg\,min}_{\mathbf{m}\in\mathcal{B}_t} D_\Psi(\mathbf{m}\|\mathbf{m}_t).$$

Then

$$\langle \boldsymbol{\phi}_t^u, \mathbf{m}_t^{\mathrm{exec}}\rangle = \min\{\langle \boldsymbol{\phi}_t^u, \mathbf{m}_t\rangle, \alpha x_t\} = u_t.$$

Proof. The online state satisfies $x_t \geq 0$, so $\mathbf{0} \in \mathcal{B}_t$. This set is compact and convex because it is the intersection of $\mathcal{M}_H$ and a closed halfspace. By (23) and the update (26)–(27), every coordinate of $\mathbf{m}_t$ is positive. By (22), $D_\Psi(\cdot\|\mathbf{m}_t)$ is continuous on $\mathcal{M}_H$. By Lemma 6, it is strictly convex and, for every $\mathbf{m} \in \mathcal{M}_H$,

$$D_\Psi(\mathbf{m}\|\mathbf{m}_t) \geq \frac{1}{2}\|\mathbf{m}-\mathbf{m}_t\|_1^2 \geq 0.$$

Thus, the projection exists and is unique, and equality holds only at $\mathbf{m} = \mathbf{m}_t$.

If

$$\langle \boldsymbol{\phi}_t^u, \mathbf{m}_t\rangle \leq \alpha x_t,$$

then $\mathbf{m}_t \in \mathcal{B}_t$ and $\mathbf{m}_t^{\mathrm{exec}} = \mathbf{m}_t$.

Now suppose

$$\langle \boldsymbol{\phi}_t^u, \mathbf{m}_t\rangle > \alpha x_t.$$

Since $\mathbf{m}_t^{\mathrm{exec}} \in \mathcal{B}_t$,

$$\langle \boldsymbol{\phi}_t^u, \mathbf{m}_t^{\mathrm{exec}}\rangle \leq \alpha x_t < \langle \boldsymbol{\phi}_t^u, \mathbf{m}_t\rangle.$$

Assume, for contradiction, that the action constraint is slack:

$$\langle \boldsymbol{\phi}_t^u, \mathbf{m}_t^{\mathrm{exec}}\rangle < \alpha x_t < \langle \boldsymbol{\phi}_t^u, \mathbf{m}_t\rangle.$$

Choose $0 < \delta < 1$ such that

$$\delta\left(\langle \boldsymbol{\phi}_t^u, \mathbf{m}_t\rangle - \langle \boldsymbol{\phi}_t^u, \mathbf{m}_t^{\mathrm{exec}}\rangle\right) < \alpha x_t - \langle \boldsymbol{\phi}_t^u, \mathbf{m}_t^{\mathrm{exec}}\rangle.$$

Such a δ exists because both differences are positive. Set

$$\mathbf{m}_\delta \triangleq (1-\delta)\mathbf{m}_t^{\mathrm{exec}} + \delta\mathbf{m}_t.$$

By convexity, $\mathbf{m}_\delta \in \mathcal{M}_H$, and

$$\begin{aligned}&\langle \boldsymbol{\phi}_t^u, \mathbf{m}_\delta\rangle\\ &= \langle \boldsymbol{\phi}_t^u, \mathbf{m}_t^{\mathrm{exec}}\rangle + \delta\left(\langle \boldsymbol{\phi}_t^u, \mathbf{m}_t\rangle - \langle \boldsymbol{\phi}_t^u, \mathbf{m}_t^{\mathrm{exec}}\rangle\right) < \alpha x_t.\end{aligned}$$

Thus $\mathbf{m}_\delta \in \mathcal{B}_t$. Since $\mathbf{m}_t \notin \mathcal{B}_t$, $\mathbf{m}_t^{\mathrm{exec}} \neq \mathbf{m}_t$, and hence $D_\Psi(\mathbf{m}_t^{\mathrm{exec}}\|\mathbf{m}_t) > 0$. Using strict convexity and $D_\Psi(\mathbf{m}_t\|\mathbf{m}_t) = 0$, we obtain

$$\begin{aligned}D_\Psi(\mathbf{m}_\delta\|\mathbf{m}_t) &< (1-\delta)D_\Psi(\mathbf{m}_t^{\mathrm{exec}}\|\mathbf{m}_t) + \delta D_\Psi(\mathbf{m}_t\|\mathbf{m}_t)\\ &= (1-\delta)D_\Psi(\mathbf{m}_t^{\mathrm{exec}}\|\mathbf{m}_t) < D_\Psi(\mathbf{m}_t^{\mathrm{exec}}\|\mathbf{m}_t),\end{aligned}$$

contradicting the definition of $\mathbf{m}_t^{\mathrm{exec}}$. Therefore, the constraint is active:

$$\langle \boldsymbol{\phi}_t^u, \mathbf{m}_t^{\mathrm{exec}}\rangle = \alpha x_t.$$

Combining the two cases with (24) proves the result. □

Lemma 8 (Closed-form solution of the OMD update)**.** *Fix $\mathbf{m}_t \in \mathcal{M}_H \cap \mathbb{R}^H_{>0}$, $\eta > 0$, and $\mathbf{g}_t \in \mathbb{R}^H$. Define $\widetilde{m}_{t+1,i} \triangleq m_{t,i}\exp(-\eta g_{t,i})$ for $i \in [H]$. Then the OMD update* (25) *has the unique solution*

$$\mathbf{m}_{t+1} = \frac{\widetilde{\mathbf{m}}_{t+1}}{\max\{1, \|\widetilde{\mathbf{m}}_{t+1}\|_1\}}.$$

Moreover, $\mathbf{m}_{t+1} \in \mathcal{M}_H \cap \mathbb{R}^H_{>0}$. Hence, the initialization (23) *keeps every OMD iterate strictly positive.*

Proof. Since $m_{t,i} > 0$, we have $\widetilde{m}_{t+1,i} > 0$ for every $i \in [H]$. For any $\mathbf{m} \in \mathcal{M}_H$,

$$m_i\log\frac{m_i}{\widetilde{m}_{t+1,i}} = m_i\log\frac{m_i}{m_{t,i}} + \eta g_{t,i}m_i,$$

where the identity at $m_i = 0$ uses $0\log 0 = 0$. Substituting into (22) gives

$$\begin{aligned}&\eta\langle \mathbf{g}_t, \mathbf{m}\rangle + D_\Psi(\mathbf{m}\|\mathbf{m}_t)\\ &= \sum_{i=1}^H\left(\eta g_{t,i}m_i + m_i\log\frac{m_i}{m_{t,i}} - m_i + m_{t,i}\right)\\ &= \sum_{i=1}^H\left(m_i\log\frac{m_i}{\widetilde{m}_{t+1,i}} - m_i + m_{t,i}\right)\\ &= D_\Psi(\mathbf{m}\|\widetilde{\mathbf{m}}_{t+1}) + \sum_{i=1}^H(m_{t,i} - \widetilde{m}_{t+1,i}).\end{aligned}$$

Thus, it suffices to minimize $D_\Psi(\mathbf{m}\|\widetilde{\mathbf{m}}_{t+1})$ over $\mathcal{M}_H$.

If $\|\widetilde{\mathbf{m}}_{t+1}\|_1 \le 1$, then $\widetilde{\mathbf{m}}_{t+1} \in \mathcal{M}_H$. Lemma 6 gives

$$D_\Psi(\mathbf{m}\|\widetilde{\mathbf{m}}_{t+1}) \ge \frac{1}{2}\|\mathbf{m}-\widetilde{\mathbf{m}}_{t+1}\|_1^2 \ge 0.$$

Since $D_\Psi(\widetilde{\mathbf{m}}_{t+1}\|\widetilde{\mathbf{m}}_{t+1}) = 0$, $\widetilde{\mathbf{m}}_{t+1}$ is the unique minimizer.

If $\|\widetilde{\mathbf{m}}_{t+1}\|_1 > 1$, set

$$\mathbf{m}^\star \triangleq \frac{\widetilde{\mathbf{m}}_{t+1}}{\|\widetilde{\mathbf{m}}_{t+1}\|_1}.$$

Then $\mathbf{m}^\star > 0$ coordinatewise, $\|\mathbf{m}^\star\|_1 = 1$, and

$$\widetilde{m}_{t+1,i} = \|\widetilde{\mathbf{m}}_{t+1}\|_1 m_i^\star.$$

Thus, for every $i \in [H]$,

$$m_i \log \frac{m_i}{\widetilde{m}_{t+1,i}} = m_i \log \frac{m_i}{m_i^\star} - m_i \log \|\widetilde{\mathbf{m}}_{t+1}\|_1,$$
$$m_i^\star \log \frac{m_i^\star}{\widetilde{m}_{t+1,i}} = -m_i^\star \log \|\widetilde{\mathbf{m}}_{t+1}\|_1.$$

Using these identities in (22) gives, for every $\mathbf{m} \in \mathcal{M}_H$,

$$\begin{aligned} &D_\Psi(\mathbf{m}\|\widetilde{\mathbf{m}}_{t+1}) - D_\Psi(\mathbf{m}^\star\|\widetilde{\mathbf{m}}_{t+1}) \\ &= \sum_{i=1}^H \left(m_i \log \frac{m_i}{m_i^\star} - m_i + m_i^\star \right) \\ &\quad + \sum_{i=1}^H (m_i^\star - m_i) \log \|\widetilde{\mathbf{m}}_{t+1}\|_1 \\ &= D_\Psi(\mathbf{m}\|\mathbf{m}^\star) + \big(1 - \|\mathbf{m}\|_1\big) \log \|\widetilde{\mathbf{m}}_{t+1}\|_1 \ge 0. \end{aligned}$$

Since $\|\mathbf{m}\|_1 \le 1$ and $\log \|\widetilde{\mathbf{m}}_{t+1}\|_1 > 0$, Lemma 6 gives the last inequality, with equality only at $\mathbf{m} = \mathbf{m}^\star$. Thus, $\mathbf{m}^\star$ is the unique minimizer.

In both cases, the minimizer has the stated formula and is strictly positive. Induction from (23) proves positivity of every iterate. □

APPENDIX C
PROOFS FOR THE REGRET UPPER BOUNDS

Lemma 9 (One-step OMD movement bound)**.** *The OMD update* (25) *with step size $\eta > 0$ satisfies*

$$\|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1 \le \eta \|\mathbf{g}_t\|_\infty \le \eta G_\alpha.$$

Proof. By Lemma 8, both iterates are positive. Applying Lemma 6 in both directions gives

$$D_\Psi(\mathbf{m}_{t+1}\|\mathbf{m}_t) \ge \frac{1}{2}\|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1^2,$$
$$D_\Psi(\mathbf{m}_t\|\mathbf{m}_{t+1}) \ge \frac{1}{2}\|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1^2.$$

Adding these bounds and using the symmetric Bregman identity,

$$\begin{aligned} &\|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1^2 \\ &\quad \le D_\Psi(\mathbf{m}_{t+1}\|\mathbf{m}_t) + D_\Psi(\mathbf{m}_t\|\mathbf{m}_{t+1}) \\ &\quad = \big\langle \nabla\Psi(\mathbf{m}_{t+1}) - \nabla\Psi(\mathbf{m}_t), \mathbf{m}_{t+1} - \mathbf{m}_t \big\rangle \\ &\quad \le \eta \langle \mathbf{g}_t, \mathbf{m}_t - \mathbf{m}_{t+1} \rangle \\ &\quad \le \eta \|\mathbf{g}_t\|_\infty \|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1. \end{aligned}$$

The last two inequalities use the first-order optimality condition for (25) with comparison point $\mathbf{m}_t$ and Hölder's inequality, respectively. If the iterates differ, divide by $\|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1$; otherwise the bound is trivial. Finally, $\|\mathbf{g}_t\|_\infty \le G_\alpha$ gives the stated uniform bound. □

Lemma 10 (OMD regret for the surrogate costs)**.** *For every $\mathbf{m} \in \mathcal{M}_H$, the update* (25) *satisfies*

$$\sum_{t=1}^T \ell_t(\mathbf{m}_t) - \sum_{t=1}^T \ell_t(\mathbf{m}) \le \frac{D_\Psi(\mathbf{m}\|\mathbf{m}_1)}{\eta} + \frac{\eta}{2}\sum_{t=1}^T \|\mathbf{g}_t\|_\infty^2. \quad (37)$$

Proof. Convexity of ℓ_t gives

$$\begin{aligned} \ell_t(\mathbf{m}_t) - \ell_t(\mathbf{m}) &\le \langle \mathbf{g}_t, \mathbf{m}_t - \mathbf{m} \rangle = \langle \mathbf{g}_t, \mathbf{m}_t - \mathbf{m}_{t+1} \rangle \\ &\quad + \langle \mathbf{g}_t, \mathbf{m}_{t+1} - \mathbf{m} \rangle. \end{aligned} \quad (38)$$

The first-order optimality condition for (25), with comparison point $\mathbf{m} \in \mathcal{M}_H$, gives

$$\begin{aligned} \eta \langle \mathbf{g}_t, \mathbf{m}_{t+1} - \mathbf{m} \rangle &\le \langle \nabla\Psi(\mathbf{m}_{t+1}) - \nabla\Psi(\mathbf{m}_t), \mathbf{m} - \mathbf{m}_{t+1} \rangle \\ &= D_\Psi(\mathbf{m}\|\mathbf{m}_t) - D_\Psi(\mathbf{m}\|\mathbf{m}_{t+1}) \\ &\quad - D_\Psi(\mathbf{m}_{t+1}\|\mathbf{m}_t). \end{aligned} \quad (39)$$

By the three-point Bregman identity, substituting (39) into (38) gives

$$\begin{aligned} \ell_t(\mathbf{m}_t) - \ell_t(\mathbf{m}) &\le \frac{D_\Psi(\mathbf{m}\|\mathbf{m}_t)}{\eta} - \frac{D_\Psi(\mathbf{m}\|\mathbf{m}_{t+1})}{\eta} \\ &\quad + \langle \mathbf{g}_t, \mathbf{m}_t - \mathbf{m}_{t+1} \rangle \\ &\quad - \frac{D_\Psi(\mathbf{m}_{t+1}\|\mathbf{m}_t)}{\eta}. \end{aligned} \quad (40)$$

Lemma 6 and Hölder's inequality give

$$\begin{aligned} &\langle \mathbf{g}_t, \mathbf{m}_t - \mathbf{m}_{t+1} \rangle - \frac{D_\Psi(\mathbf{m}_{t+1}\|\mathbf{m}_t)}{\eta} \\ &\quad \le \|\mathbf{g}_t\|_\infty \|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1 - \frac{\|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1^2}{2\eta} \\ &\quad \le \frac{\eta}{2}\|\mathbf{g}_t\|_\infty^2. \end{aligned} \quad (41)$$

The last step is Young's inequality. Substituting (41) into (40) and summing over t gives

$$\begin{aligned} \sum_{t=1}^T \ell_t(\mathbf{m}_t) - \sum_{t=1}^T \ell_t(\mathbf{m}) &\le \frac{D_\Psi(\mathbf{m}\|\mathbf{m}_1)}{\eta} - \frac{D_\Psi(\mathbf{m}\|\mathbf{m}_{T+1})}{\eta} \\ &\quad + \frac{\eta}{2}\sum_{t=1}^T \|\mathbf{g}_t\|_\infty^2. \end{aligned}$$

Since $D_\Psi(\mathbf{m}\|\mathbf{m}_{T+1}) \ge 0$, this proves (37). □

Lemma 11 (Positive-part representation of the dynamics)**.** *For $t \in [T]$, $x \ge 0$, and $\mathbf{m} \in \mathcal{M}_H$, define*

$$F_t(x; \mathbf{m}) \triangleq [\alpha x - \langle \boldsymbol{\phi}_t^u, \mathbf{m} \rangle]_+ + w_t. \quad (42)$$

Then, for every $t \in [T]$ and $\mathbf{m} \in \mathcal{M}_H$,

$$x_{t+1} = F_t(x_t; \mathbf{m}_t), \qquad x_{t+1}(\mathbf{m}) = F_t(x_t(\mathbf{m}); \mathbf{m}). \quad (43)$$

Moreover, for all $x, x' \ge 0$ and $\mathbf{m}, \mathbf{m}' \in \mathcal{M}_H$,

$$|F_t(x; \mathbf{m}) - F_t(x'; \mathbf{m}')| \le \alpha|x - x'| + \|\boldsymbol{\phi}_t^u\|_\infty \|\mathbf{m} - \mathbf{m}'\|_1. \quad (44)$$

Proof. By (3) and (24),

$$\begin{aligned} x_{t+1} &= \alpha x_t - \min\{\langle \boldsymbol{\phi}_t^u, \mathbf{m}_t\rangle, \alpha x_t\} + w_t \\ &= [\alpha x_t - \langle \boldsymbol{\phi}_t^u, \mathbf{m}_t\rangle]_+ + w_t = F_t(x_t; \mathbf{m}_t). \end{aligned}$$

For a fixed $\mathbf{m}$, (11) and Lemma 2 give $\langle \boldsymbol{\phi}_t^u, \mathbf{m}\rangle = u_t(\mathbf{m}) \le \alpha x_t(\mathbf{m})$. Hence, by (3),

$$\begin{aligned} x_{t+1}(\mathbf{m}) &= \alpha x_t(\mathbf{m}) - \langle \boldsymbol{\phi}_t^u, \mathbf{m}\rangle + w_t \\ &= [\alpha x_t(\mathbf{m}) - \langle \boldsymbol{\phi}_t^u, \mathbf{m}\rangle]_+ + w_t = F_t(x_t(\mathbf{m}); \mathbf{m}). \end{aligned}$$

Finally, since $|[z]_+ - [z']_+| \le |z - z'|$,

$$\begin{aligned} &|F_t(x; \mathbf{m}) - F_t(x'; \mathbf{m}')| \\ &= \big|[\alpha x - \langle \boldsymbol{\phi}_t^u, \mathbf{m}\rangle]_+ - [\alpha x' - \langle \boldsymbol{\phi}_t^u, \mathbf{m}'\rangle]_+\big| \\ &\le |\alpha(x - x') - \langle \boldsymbol{\phi}_t^u, \mathbf{m} - \mathbf{m}'\rangle| \\ &\le \alpha|x - x'| + \|\boldsymbol{\phi}_t^u\|_\infty \|\mathbf{m} - \mathbf{m}'\|_1, \end{aligned}$$

where the last inequality follows from the triangle and Hölder inequalities. □

Proof of Lemma 5. Let $d_t \triangleq |x_t - x_t(\mathbf{m}_t)|$. Since both trajectories start at zero, $d_1 = 0$. For $1 \le t < T$, Lemma 11, applied with the same parameter $\mathbf{m}_t$, gives

$$\begin{aligned} |x_{t+1} - x_{t+1}(\mathbf{m}_t)| &= |F_t(x_t; \mathbf{m}_t) - F_t(x_t(\mathbf{m}_t); \mathbf{m}_t)| \\ &\le \alpha d_t. \end{aligned}$$

The affine state formula (34), the coefficient bound (35), and Lemma 9 give

$$\begin{aligned} &|x_{t+1}(\mathbf{m}_t) - x_{t+1}(\mathbf{m}_{t+1})| \\ &\quad = |\langle \boldsymbol{\phi}_{t+1}^x, \mathbf{m}_t - \mathbf{m}_{t+1}\rangle| \\ &\quad \le \|\boldsymbol{\phi}_{t+1}^x\|_\infty \|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1 \\ &\quad \le \frac{\alpha\eta G_\alpha}{1-\alpha}. \end{aligned}$$

Thus, the triangle inequality yields

$$\begin{aligned} d_{t+1} &\le |x_{t+1} - x_{t+1}(\mathbf{m}_t)| + |x_{t+1}(\mathbf{m}_t) - x_{t+1}(\mathbf{m}_{t+1})| \\ &\le \alpha d_t + \frac{\alpha\eta G_\alpha}{1-\alpha}. \end{aligned}$$

Iterating from $d_1 = 0$ gives, for $2 \le t \le T$,

$$d_t \le \frac{\alpha\eta G_\alpha}{1-\alpha} \sum_{j=0}^{t-2} \alpha^j \le \frac{\alpha\eta G_\alpha}{1-\alpha} \sum_{j=0}^{\infty} \alpha^j = \frac{\alpha\eta G_\alpha}{(1-\alpha)^2}.$$

This bound also holds at $t = 1$.

The clipping rule (24) and feasibility of each fixed policy (Lemma 2) give

$$\begin{aligned} 0 \le u_t(\mathbf{m}_t) - u_t &= [u_t(\mathbf{m}_t) - \alpha x_t]_+ \le [\alpha x_t(\mathbf{m}_t) - \alpha x_t]_+ \\ &\le \alpha d_t. \end{aligned}$$

Finally, $\ell_t(\mathbf{m}_t) = c_t(x_t(\mathbf{m}_t), u_t(\mathbf{m}_t))$. Both state–action pairs lie in $\mathcal{D}_\alpha$, so the Lipschitz bound (5) gives

$$\begin{aligned} c_t(x_t, u_t) - \ell_t(\mathbf{m}_t) &\le L_x d_t + L_u |u_t - u_t(\mathbf{m}_t)| \\ &\le (L_x + \alpha L_u) d_t \le \frac{\alpha\eta G_\alpha (L_x + \alpha L_u)}{(1-\alpha)^2}. \end{aligned}$$

Summing over $t = 1, \dots, T$ proves (30). □

Proof of Theorem 1. Fix $\mathbf{m} \in \mathcal{M}_H$. We first bound the initial Bregman divergence. Since $\mathbf{m}_1 = \frac{1}{H+1}\mathbf{1}$ and $m_i \log m_i \le 0$ for $m_i \in [0, 1]$,

$$\begin{aligned} D_\Psi(\mathbf{m}\|\mathbf{m}_1) &= \sum_{i=1}^{H} m_i \log m_i \\ &\quad + \|\mathbf{m}\|_1 \big(\log(H+1) - 1\big) + \frac{H}{H+1} \\ &\le (1 - \|\mathbf{m}\|_1) \frac{H}{H+1} \\ &\quad + \|\mathbf{m}\|_1 \left(\log(H+1) - \frac{1}{H+1}\right) \\ &\le \log(H+1). \end{aligned}$$

The last line uses $\|\mathbf{m}\|_1 \in [0, 1]$ and $\frac{H}{H+1} \le \log(H+1)$, which follows from $\log z \ge 1 - z^{-1}$ for $z \ge 1$.

Lemma 10 and $\|\mathbf{g}_t\|_\infty \le G_\alpha$ now give

$$\begin{aligned} \sum_{t=1}^{T} \big(\ell_t(\mathbf{m}_t) - \ell_t(\mathbf{m})\big) &\le \frac{D_\Psi(\mathbf{m}\|\mathbf{m}_1)}{\eta} + \frac{\eta}{2} \sum_{t=1}^{T} \|\mathbf{g}_t\|_\infty^2 \\ &\le \frac{\log(H+1)}{\eta} + \frac{\eta T G_\alpha^2}{2}. \end{aligned}$$

Since $\ell_t(\mathbf{m}) = c_t(x_t(\mathbf{m}), u_t(\mathbf{m}))$, adding the stability bound (30) yields

$$\begin{aligned} J_T^{\mathrm{on}} - J_T(\mathbf{m}) &= \sum_{t=1}^{T} \big(c_t(x_t, u_t) - \ell_t(\mathbf{m}_t)\big) \\ &\quad + \sum_{t=1}^{T} \big(\ell_t(\mathbf{m}_t) - \ell_t(\mathbf{m})\big) \\ &\le \frac{\log(H+1)}{\eta} + \frac{\eta T G_\alpha^2}{2} \\ &\quad + \frac{\alpha\eta T G_\alpha (L_x + \alpha L_u)}{(1-\alpha)^2} \\ &= \frac{\log(H+1)}{\eta} + \eta T \Gamma_\alpha. \end{aligned}$$

Taking the maximum over $\mathbf{m} \in \mathcal{M}_H$ proves (28). At the stated step size,

$$\frac{\log(H+1)}{\eta^*} = \eta^* T \Gamma_\alpha = \sqrt{\Gamma_\alpha T \log(H+1)},$$

so the two terms sum to the claimed regret bound.

Finally, for fixed $L_x, L_u > 0$, as $\alpha \uparrow 1$,

$$(1-\alpha) G_\alpha = \alpha\big(L_x + (1-\alpha) L_u\big) \longrightarrow L_x,$$

and hence

$$\begin{aligned} (1-\alpha)^3 \Gamma_\alpha &= \frac{1-\alpha}{2} \big((1-\alpha) G_\alpha\big)^2 \\ &\quad + \alpha\big((1-\alpha) G_\alpha\big)(L_x + \alpha L_u) \\ &\longrightarrow L_x (L_x + L_u) > 0. \end{aligned}$$

Thus $\sqrt{\Gamma_\alpha} = \Theta((1-\alpha)^{-3/2})$. □

Proof of Corollary 2. Fix $\mathbf{q} \in \mathcal{Q}_\alpha$ and set $\mathbf{m} = \mathbf{m}_{H_T}(\mathbf{q}) \in \mathcal{M}_{H_T}$ as in Lemma 4. By the definition of $\mathrm{Reg}_T^{\mathrm{SDAC}}$,

$$\begin{aligned} J_T^{\mathrm{on}} - J_T(\mathbf{q}) &= J_T^{\mathrm{on}} - J_T(\mathbf{m}) + J_T(\mathbf{m}) - J_T(\mathbf{q}) \\ &\le \mathrm{Reg}_T^{\mathrm{SDAC}} + J_T(\mathbf{m}) - J_T(\mathbf{q}). \end{aligned}$$

The approximation bound (19) gives

$$J_T(\mathbf{m}) - J_T(\mathbf{q}) \le \left(L_u + \frac{L_x}{1-\alpha}\right)\alpha^{H_T+1}T.$$

This bound is independent of $\mathbf{q}$. Taking the supremum and applying Theorem 1 with step size η^* therefore yields

$$\begin{aligned}\mathrm{Reg}_T^{\mathrm{alloc}} &= \sup_{\mathbf{q}\in\mathcal{Q}_\alpha}\left(J_T^{\mathrm{on}} - J_T(\mathbf{q})\right)\\ &\le \mathrm{Reg}_T^{\mathrm{SDAC}} + \left(L_u + \frac{L_x}{1-\alpha}\right)\alpha^{H_T+1}T\\ &\le 2\sqrt{\Gamma_\alpha T\log(H_T+1)}\\ &\quad + \left(L_u + \frac{L_x}{1-\alpha}\right)\alpha^{H_T+1}T,\end{aligned}$$

which proves (31).

For the rate, the definition of H_T gives

$$\frac{\log T}{\log(1/\alpha)} \le H_T \le 1 + \frac{\log T}{\log(1/\alpha)}.$$

Since $0<\alpha<1$, it follows that

$$\begin{aligned}\alpha^{H_T+1}T &\le \alpha^{1+\log T/\log(1/\alpha)}T = \alpha\exp\left(\frac{\log\alpha}{\log(1/\alpha)}\log T\right)T\\ &= \alpha e^{-\log T}T = \alpha.\end{aligned}$$

Also, for fixed α,

$$\log(H_T+1) \le \log\left(2 + \frac{\log T}{\log(1/\alpha)}\right) = O(\log\log T).$$

Thus, for fixed α, L_x, L_u, the right-hand side of (31) is $O(\sqrt{T\log\log T}) = o(T)$. □

Proof of Corollary 3. By Corollary 1, for every $k\in[0,\alpha]$,

$$\mathbf{q}(k)\in\mathcal{Q}_\alpha, \qquad J_T(\mathbf{q}(k)) = J_T(k).$$

Hence, by Corollary 2,

$$\begin{aligned}\mathrm{Reg}_T^{\mathrm{frac}} &= J_T^{\mathrm{on}} - \min_{k\in[0,\alpha]} J_T(k) = J_T^{\mathrm{on}} - \min_{k\in[0,\alpha]} J_T(\mathbf{q}(k))\\ &\le J_T^{\mathrm{on}} - \inf_{\mathbf{q}\in\mathcal{Q}_\alpha} J_T(\mathbf{q}) = \mathrm{Reg}_T^{\mathrm{alloc}}\\ &\le 2\sqrt{\Gamma_\alpha T\log(H_T+1)}\\ &\quad + \left(L_u + \frac{L_x}{1-\alpha}\right)\alpha^{H_T+1}T.\end{aligned}$$

The rate follows from the same corollary. □

Appendix D
Proofs for the Minimax Regret Lower Bound

Proof of Theorem 2. Fix a causal feasible controller $\mathcal{A}$. Let $\xi_1,\dots,\xi_T$ be independent Rademacher random variables, independent of the controller's randomization. Set

$$w_t = 1, \qquad c_t(x,u) = \xi_t(L_x x - L_u u), \qquad t\in[T].$$

Each cost is affine with gradient $\xi_t(L_x, -L_u)$, so it is convex and satisfies the required subgradient bounds. Also,

$$\begin{aligned}|c_t(x,u) - c_t(x',u')| &= |L_x(x-x') - L_u(u-u')|\\ &\le L_x|x-x'| + L_u|u-u'|.\end{aligned}$$

Thus every realization gives an admissible sequence.

Consider the two fixed SDAC parameters

$$\mathbf{m}^{(0)} = \mathbf{0}, \qquad \mathbf{m}^{(1)} = \mathbf{e}_1 = (1,0,\dots,0)\in\mathcal{M}_H.$$

Both trajectories start at $(x_1,u_1) = (0,0)$. For $t\ge 2$, the SDAC action formula and the dynamics give

$$\begin{aligned}&u_t(\mathbf{m}^{(0)}) = 0, \quad x_t(\mathbf{m}^{(0)}) = \sum_{s=0}^{t-2}\alpha^s = \frac{1-\alpha^{t-1}}{1-\alpha},\\ &u_t(\mathbf{m}^{(1)}) = \alpha, \quad x_t(\mathbf{m}^{(1)}) = 1.\end{aligned}$$

The last identity follows from $x_2(\mathbf{m}^{(1)}) = 1$ and

$$x_{t+1}(\mathbf{m}^{(1)}) = \alpha x_t(\mathbf{m}^{(1)}) - \alpha + 1 = \alpha - \alpha + 1 = 1$$

whenever $x_t(\mathbf{m}^{(1)}) = 1$. Hence, for $t\ge 2$,

$$x_t(\mathbf{m}^{(0)}) - x_t(\mathbf{m}^{(1)}) = \frac{1-\alpha^{t-1}}{1-\alpha} - 1 = \frac{\alpha(1-\alpha^{t-2})}{1-\alpha}.$$

Since both policies incur zero cost at $t=1$, their cumulative costs satisfy

$$\begin{aligned}J_T(\mathbf{m}^{(0)}) - J_T(\mathbf{m}^{(1)}) &= \sum_{t=2}^{T}\xi_t\left(\frac{\alpha L_x(1-\alpha^{t-2})}{1-\alpha} + \alpha L_u\right)\\ &= \sum_{t=2}^{T}\xi_t\beta_t.\end{aligned}$$

By causality, (x_t,u_t) depends only on past signs and the controller's randomization, so it is independent of ξ_t. Since $\mathbb{E}[\xi_t]=0$,

$$\mathbb{E}_{\boldsymbol{\xi},\mathcal{A}}[J_T^{\mathrm{on}}] = \sum_{t=1}^{T}\mathbb{E}[\xi_t]\,\mathbb{E}_{\boldsymbol{\xi},\mathcal{A}}[L_x x_t - L_u u_t] = 0.$$

The two fixed-policy trajectories are deterministic. Thus, for $j\in\{0,1\}$,

$$\begin{aligned}\mathbb{E}_{\boldsymbol{\xi}}[J_T(\mathbf{m}^{(j)})] &= \sum_{t=1}^{T}\mathbb{E}[\xi_t]\left(L_x x_t(\mathbf{m}^{(j)}) - L_u u_t(\mathbf{m}^{(j)})\right)\\ &= 0.\end{aligned}$$

Both parameters belong to $\mathcal{M}_H$. Using $\min\{a,b\} = (a+b-|a-b|)/2$, we obtain

$$\begin{aligned}\mathrm{Reg}_T^{\mathrm{SDAC}} &= J_T^{\mathrm{on}} - \min_{\mathbf{m}\in\mathcal{M}_H} J_T(\mathbf{m})\\ &\ge J_T^{\mathrm{on}} - \min\{J_T(\mathbf{m}^{(0)}), J_T(\mathbf{m}^{(1)})\}\\ &= J_T^{\mathrm{on}} - \frac{J_T(\mathbf{m}^{(0)}) + J_T(\mathbf{m}^{(1)})}{2}\\ &\quad + \frac{1}{2}\left|J_T(\mathbf{m}^{(0)}) - J_T(\mathbf{m}^{(1)})\right|.\end{aligned}$$

Taking expectations and applying Khintchine's inequality in the last line gives

$$\begin{aligned}\mathbb{E}_{\boldsymbol{\xi},\mathcal{A}}[\mathrm{Reg}_T^{\mathrm{SDAC}}] &\ge \frac{1}{2}\mathbb{E}_{\boldsymbol{\xi}}\left|J_T(\mathbf{m}^{(0)}) - J_T(\mathbf{m}^{(1)})\right|\\ &= \frac{1}{2}\mathbb{E}_{\boldsymbol{\xi}}\left|\sum_{t=2}^{T}\xi_t\beta_t\right| \ge \frac{1}{2\sqrt{2}}\sqrt{\sum_{t=2}^{T}\beta_t^2}.\end{aligned}$$

Averaging over $\boldsymbol{\xi}\in\{-1,1\}^T$, some $\boldsymbol{\xi}^\star$ gives expected regret at least this bound over the controller's randomization. Fixing

$\boldsymbol{\xi}^\star$ before the interaction gives the required deterministic oblivious sequence. □

Proof of Corollary 4. For $t = t_{\mathrm{mix}}(\alpha) + 2, \dots, T$,

$$\alpha^{t-2} \le \alpha^{t_{\mathrm{mix}}(\alpha)} \le \alpha^{\log 2/\log(1/\alpha)} = \frac{1}{2}.$$

Hence, by (33),

$$\beta_t \ge \alpha\left(L_u + \frac{L_x}{2(1-\alpha)}\right) \ge \frac{\alpha}{2}\left(L_u + \frac{L_x}{1-\alpha}\right) = \frac{G_\alpha}{2}.$$

There are $T - t_{\mathrm{mix}}(\alpha) - 1 \ge T/2$ such indices, since $T \ge 2t_{\mathrm{mix}}(\alpha) + 2$. Theorem 2 therefore gives

$$\begin{aligned}
\mathbb{E}_{\mathcal{A}}\left[\mathrm{Reg}_T^{\mathrm{SDAC}}\right] &\ge \frac{1}{2\sqrt{2}}\sqrt{\sum_{t=2}^{T}\beta_t^2} \\
&\ge \frac{1}{2\sqrt{2}}\left[\left(T - t_{\mathrm{mix}}(\alpha) - 1\right)\left(\frac{G_\alpha}{2}\right)^2\right]^{1/2} \\
&\ge \frac{1}{2\sqrt{2}}\left[\frac{T}{2}\left(\frac{G_\alpha}{2}\right)^2\right]^{1/2} = \frac{G_\alpha}{8}\sqrt{T}. \qquad \square
\end{aligned}$$

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